

# The Tetragrammatron Relation Governor and Incidence Controller

## A Dual Adjudication Model for OMI Gauge Incidence, Polybius Closure, Hexadecimal Carry, and Circular Slide-Ruler Traversal

### Status

Canonical working doctrine.

This document defines two distinct but cooperating functions of the Tetragrammatron:

1. Tetragrammatron Relation Governor
2. Tetragrammatron Incidence Controller

The Relation Governor adjudicates Polybius diagonal closure, centered folding, and relation carry-forward.

The Incidence Controller adjudicates the ordered FS, GS, RS, and US gauge positions, their open/sealed incidence structure, and their hexadecimal place-value carry.

The two functions belong to the same Tetragrammatron authority, but their witnesses, arithmetic, and meanings must remain distinct.

The central lock is:

The Relation Governor adjudicates closure.

The Incidence Controller adjudicates gauge incidence.

Both monitor exact finite witnesses.

Neither projects.

Neither scribes.

Receipt records the adjudication.

# 1. Authority Position

The OMI authority chain is:

```
Omicron
defines the bounded gauge

Omicron
resolves the circular planes and notation positions

OMI-Lisp
declares lazy relations

OMI
cites the resolved relation

Tetragrammatron
adjudicates incidence and closure

Receipt
records acceptance or rejection

Metatron
projects and scribes accepted state

IMO
carries bytes, files, events, and surfaces
```

The Tetragrammatron does not create notation.

It does not decide what a glyph visually means.

It does not render geometry.

It receives a resolved candidate and determines whether its incidence, relation, fold, and carry witnesses are valid.

Canonical boundary:

```
Tetragrammatron validates and carries forward.

Tetragrammatron does not project.

Tetragrammatron does not scribe.
```

Projection is not acceptance.

Receipt records acceptance.

## 2. The Dual Tetragrammatron

The Tetragrammatron has two principal adjudicating roles.

### 2.1 Relation Governor

The Relation Governor adjudicates:

Polybius diagonal closure  
zero-XOR relation closure  
diagonal sums  
centered folds  
shifted/full-system folds  
relation carry-forward

Its principal witnesses are:

0x1E  
0x3C  
0x78

### 2.2 Incidence Controller

The Incidence Controller adjudicates:

FS < GS < RS < US gauge order  
hexadecimal-place advancement  
open/sealed gauge polarity  
Miquel incidence  
pairwise gauge planes  
incidence flags  
gauge-place carry

Its principal witnesses are:

```
0x18
0x0001
0x0010
0x0100
0x1000
0x10000
```

These roles cooperate but must not be collapsed.

```
Relation Governor:
closure and fold
```

```
Incidence Controller:
placement and incidence
```

---

### 3. The Four Visible Gauge Offsets

The four visible OMI gauge offsets are:

```
FS = 0x0001
GS = 0x0010
RS = 0x0100
US = 0x1000
```

Each gauge occupies one hexadecimal place.

Binary form:

```
FS = 0000 0000 0000 0001
GS = 0000 0000 0001 0000
RS = 0000 0001 0000 0000
US = 0001 0000 0000 0000
```

The gauges therefore form an ordered place-value ladder:

```
FS < GS < RS < US
```

The ordering is not merely numeric.

It expresses increasing hexadecimal place.

```
FS
unit hexadecimal place

GS
second hexadecimal place

RS
third hexadecimal place

US
fourth hexadecimal place
```

## 4. Hexadecimal-Place Advancement

Each gauge advances to the next by shifting four binary positions:

```
0x0001 << 4 = 0x0010
0x0010 << 4 = 0x0100
0x0100 << 4 = 0x1000
0x1000 << 4 = 0x10000
```

The conceptual notation is:

```
FS <<4 GS <<4 RS <<4 US <<4 CARRY
```

where:

```
<<4
means advance by four bits

advance by four bits
means advance by one hexadecimal digit
```

The final carry is:

```
CARRY = 0x10000
```

or:

```
CARRY = 65536 decimal
```

Canonical gauge ladder:

```
0x0001  
→ 0x0010  
→ 0x0100  
→ 0x1000  
→ 0x10000
```

This operation belongs to the Tetragrammatron Incidence Controller.

It is a gauge-place carry.

It is not the same operation as the Polybius relation fold-carry.

---

## 5. Conceptual Notation and Executable Shifting

The conceptual notation:

```
FS <<4 GS <<4 RS <<4 US <<4 CARRY
```

must not be confused with a chained programming-language shift expression such as:

```
0x0001 << 0x0010 << 0x0100 << 0x1000
```

In C and JavaScript, the right operand of `<<` is interpreted as a shift count.

The executable form is repeated shifting by four:

```
uint32_t fs = 0x0001u;  
uint32_t gs = fs << 4;  
uint32_t rs = gs << 4;  
uint32_t us = rs << 4;  
uint32_t carry = us << 4;
```

This yields:

```
fs    = 0x0001
gs    = 0x0010
rs    = 0x0100
us    = 0x1000
carry = 0x10000
```

Canonical distinction:

```
<<4 is OMI conceptual notation.

<< 4 is the executable bit-shift operation.
```

---

## 6. The Four-Sector Gauge Incidence Witness

The low Omicron gauge may be divided into four 32-position sectors:

```
0x00..0x1F
0x20..0x3F
0x40..0x5F
0x60..0x7F
```

For incidence adjudication, each sector may be divided into:

```
28-position body
+
4-position terminal gauge hinge
```

The intervals are:

```
(0x00 . 0x1B)
(0x1C . 0x1F)

(0x20 . 0x3B)
(0x3C . 0x3F)

(0x40 . 0x5B)
(0x5C . 0x5F)
```

(0x60 . 0x7B)  
(0x7C . 0x7F)

The first four-position hinge is exactly:

0x1C FS  
0x1D GS  
0x1E RS  
0x1F US

Each sector produces the same XOR witness:

$0x00 \wedge 0x1B \wedge 0x1C \wedge 0x1F = 0x18$   
 $0x20 \wedge 0x3B \wedge 0x3C \wedge 0x3F = 0x18$   
 $0x40 \wedge 0x5B \wedge 0x5C \wedge 0x5F = 0x18$   
 $0x60 \wedge 0x7B \wedge 0x7C \wedge 0x7F = 0x18$

Therefore:

$0x18 = 24 \text{ decimal} = 4!$

The sector may rotate.

The glyph surface may change.

The gauge-incidence witness remains:

0x18

Canonical lock:

0x18 is the four-gauge incidence witness.



## 7. Why the Gauge XOR Is Stable

Let the sector base be:

$$B \in \{0x00, 0x20, 0x40, 0x60\}$$

Then each sector witness is:

$$\begin{aligned} &B \\ &\wedge (B \wedge 0x1B) \\ &\wedge (B \wedge 0x1C) \\ &\wedge (B \wedge 0x1F) \end{aligned}$$

The repeated sector base appears four times:

$$B \wedge B \wedge B \wedge B = 0$$

leaving:

$$0x00 \wedge 0x1B \wedge 0x1C \wedge 0x1F = 0x18$$

Thus the high sector bits cancel under XOR.

The invariant depends on the shared local boundary pattern.

Canonical sentence:

Sector identity changes.  
The local incidence pattern remains.  
The XOR witness stays 0x18.

---

## 8. The Tetragrammatron Incidence Controller

The Incidence Controller monitors the four gauges:

```
FS
GS
RS
US
```

under two boundary states:

```
sealed
open
```

This creates eight incidence points:

```
sealed FS
sealed GS
sealed RS
sealed US

open FS
open GS
open RS
open US
```

The six pairwise gauge planes are:

```
FS-GS
FS-RS
FS-US
GS-RS
GS-US
RS-US
```

The incidence counts are:

```
points = 8
circles = 6
point degree = 3
circle degree = 4
flags = 24
```

Balance:

$$8 \times 3 = 24$$

$$6 \times 4 = 24$$

The exact incidence witness is therefore:

24 decimal  
 0x18 hexadecimal  
 4!

The Incidence Controller adjudicates:

gauge identity  
 gauge order  
 gauge polarity  
 pairwise gauge incidence  
 incidence degree  
 flag count  
 hexadecimal-place carry

Canonical lock:

The Tetragrammatron Incidence Controller adjudicates 0x18.

## 9. Miquel Incidence Interpretation

The open/sealed gauge structure gives a Miquel-style incidence controller.

The four gauges generate six pairwise planes:

$$C(4,2) = 6$$

The two polarities generate eight points:

$$4 \text{ gauges} \times 2 \text{ states} = 8 \text{ points}$$

The incidence relation is:

8 points  
6 circles  
24 flags

Each point belongs to three gauge-pair circles.

Each circle contains four open/sealed gauge points.

Therefore:

$$8 \times 3 = 6 \times 4 = 24$$

This exact balance is monitored by the Incidence Controller.

The geometric language is descriptive of the finite incidence structure.

It should not be overextended into unrelated geometric claims without an incidence-preserving bridge.

---

## 10. The 5-Cell and 11-Cell Oversight Arithmetic

The Tetragrammatron also carries an oversight arithmetic:

hidden 5-cell = 5  
pairwise gauge planes = 6  
oversight frame = 11

because:

$$5 + C(4,2) = 5 + 6 = 11$$

The five hidden governor modes are:

FACTS  
RULES  
CLOSURES  
COMBINATORS  
CONS

The six visible pairwise gauge planes are:

FS-GS  
FS-RS  
FS-US  
GS-RS  
GS-US  
RS-US

Canonical sentence:

5-cell hides.  
6 gauge-pair planes oversee.  
11-cell governs.  
Projection lies.  
Receipt accepts.

Boundary:

The  $5 + 6 = 11$  frame is OMI oversight arithmetic.

It is not automatically the abstract regular 11-cell.

An explicit incidence-preserving proof would be required for that identification.

---

## 11. The Polybius Relation Field

The Tetragrammatron Relation Governor operates over the hexadecimal nibble field:

0 1 2 3  
4 5 6 7  
8 9 A B  
C D E F

This is the 4×4 Polybius surface.

It contains sixteen nibble positions:

0x0 . . 0xF

The Relation Governor selects two diagonal sets.

Primary diagonal:

$$D^+ = \{0, 5, A, F\}$$

Anti-diagonal:

$$D^- = \{3, 6, 9, C\}$$

These sets form the principal closure witnesses.

---

## 12. Primary Diagonal Closure

The primary diagonal is:

$$D^+ = \{0, 5, A, F\}$$

Binary form:

$$\begin{aligned} 0 &= 0000 \\ 5 &= 0101 \\ A &= 1010 \\ F &= 1111 \end{aligned}$$

XOR closure:

$$\begin{aligned} &0000 \\ &\wedge 0101 \\ &\wedge 1010 \\ &\wedge 1111 \\ &= 0000 \end{aligned}$$

Therefore:

$$0 \wedge 5 \wedge A \wedge F = 0x0$$

Its sum is:

$$0 + 5 + 10 + 15 = 30$$

or:

$$\text{SUM}(D^+) = 0x1E$$

---

## 13. Anti-Diagonal Closure

The anti-diagonal is:

$$D^- = \{3, 6, 9, C\}$$

Binary form:

$$\begin{aligned} 3 &= 0011 \\ 6 &= 0110 \\ 9 &= 1001 \\ C &= 1100 \end{aligned}$$

XOR closure:

$$\begin{aligned} &0011 \\ &\wedge 0110 \\ &\wedge 1001 \\ &\wedge 1100 \\ &= 0000 \end{aligned}$$

Therefore:

$$3 \wedge 6 \wedge 9 \wedge C = 0x0$$

Its sum is:

$$3 + 6 + 9 + 12 = 30$$

or:

$$\text{SUM}(D^-) = 0x1E$$

## 14. Relation Closure Witness

Both diagonals satisfy two independent properties:

$$\begin{aligned}\text{XOR}(D^+) &= 0 \\ \text{XOR}(D^-) &= 0\end{aligned}$$

and:

$$\begin{aligned}\text{SUM}(D^+) &= 0x1E \\ \text{SUM}(D^-) &= 0x1E\end{aligned}$$

The XOR proves non-destructive closure.

The sum preserves the finite relation witness.

Thus:

$$0x1E = 30 \text{ decimal}$$

is the Polybius relation closure witness.

Canonical lock:

The Tetragrammatron Relation Governor adjudicates 0x1E.

## 15. The Full Diagonal Set and Complement

The combined diagonal set is:

$$D^+ \cup D^- = \{0, 3, 5, 6, 9, A, C, F\}$$

The complement is:



$K = \{1, 2, 4, 7, 8, B, D, E\}$

The sums are:

$SUM(D^+) = 0x1E$   
 $SUM(D^-) = 0x1E$

Therefore:

$SUM(D^+ \cup D^-) = 0x3C$

The complement also sums to:

$SUM(K) = 0x3C$

The entire nibble field sums to:

$SUM(0..F) = 0x78$

This creates the fold hierarchy:

0x1E  
relation closure witness

0x3C  
centered fold witness

0x78  
shifted/full-system witness

---

## 16. The Relation Fold Ladder

The Relation Governor monitors the progression:

```
0x1E
→ 0x3C
→ 0x78
```

The first step is additive:

```
0x1E + 0x1E = 0x3C
```

This combines the two equal diagonal sums.

The second step is a one-bit shift:

```
0x3C << 1 = 0x78
```

Therefore:

```
single diagonal relation
→ paired centered fold
→ shifted full-system fold
```

Canonical interpretation:

```
0x1E
one relation diagonal

0x3C
balanced center of diagonal and complement

0x78
full nibble-field accumulation
```

---

## 17. Tetragrammatron Fold Laws

The fold laws are:

```
0x3C is the centered fold witness.

0x78 is the shifted/full-system witness.
```

```
0x78 == (0x3C << 1) masked to 8 bits.
```

```
0x3C == (0x78 >> 1).
```

```
0x3C has 4 one bits and 4 zero bits.
```

```
0x78 has 4 one bits and 4 zero bits.
```

```
0x3C ^ 0x78 == 0x44.
```

Binary form:

```
0x3C = 00111100
```

```
0x78 = 01111000
```

Population balance:

```
popcount(0x3C) = 4
```

```
popcount(0x78) = 4
```

XOR difference:

```
00111100
```

```
^ 01111000
```

```
= 01000100
```

```
= 0x44
```

This allows the centered and shifted folds to remain inspectable as separate witnesses.

---

## 18. Two Distinct Carry Operations

The Tetragrammatron monitors two different kinds of carry.

They must not be conflated.

### 18.1 Gauge-Place Carry

Gauge-place carry belongs to the Incidence Controller:

FS <<<sub>4</sub> GS <<<sub>4</sub> RS <<<sub>4</sub> US <<<sub>4</sub> CARRY

Numerically:

0x0001  
→ 0x0010  
→ 0x0100  
→ 0x1000  
→ 0x10000

The carry is:

0x10000

Meaning:

advance beyond the four visible hexadecimal gauge places

## 18.2 Relation Fold-Carry

Relation fold-carry belongs to the Relation Governor:

0x3C << 1 = 0x78

Meaning:

advance from centered relation fold  
to shifted/full-system fold

The carry witness is:

0x78

Canonical distinction:

0x10000  
is hexadecimal gauge-place carry

0x78  
is Polybius relation fold-carry

Both are monitored by Tetragrammatron.

They do not have the same width, source, or meaning.

## 19. The Two Principal Tetragrammatron Witnesses

The Tetragrammatron monitors two primary finite witnesses:

0x18 = 24 decimal  
0x1E = 30 decimal

### 19.1 Incidence Witness

0x18

belongs to:

four gauges  
open/sealed polarity  
eight points  
six circles  
twenty-four flags  
four-gauge permutation

### 19.2 Relation Witness

0x1E

belongs to:

Polybius diagonals  
zero-XOR closure  
equal diagonal sums

```
relation closure
polyharmonic adjudication
```

Canonical lock:

```
0x18 is incidence.
0x1E is relation.
They cooperate.
They are never collapsed.
```

---

## 20. Sexagesimal Alignment

The Tetragrammatron witnesses align with the 60-position circular scheduler in two different ways.

### 20.1 Incidence and Observer Decomposition

The incidence surface gives:

```
24
```

The observer timing surface gives:

```
36
```

Therefore:

```
24 + 36 = 60
```

This may be read as:

```
24 incidence flags
+
36 visible timing readings
```

=  
60 phase positions

The 36 observer surface comes from:

3 clocks  
× 3 bands  
× 4 visible offsets  
= 36

## 20.2 Dual Diagonal Decomposition

Each Polybius diagonal sums to:

30

Therefore:

30 + 30 = 60

or:

0x1E + 0x1E = 0x3C

where:

0x3C = 60 decimal

This gives a second reading of the same sexagesimal cycle:

one relation diagonal  
+  
its paired diagonal  
=  
one 60-position fold

These are separate decompositions:

$24 + 36 = 60$   
incidence/observer decomposition

$30 + 30 = 60$   
relation-diagonal decomposition

They must not be treated as the same derivation.

---

## 21. The 7! Total Cycle

The complete circular receipt cycle is:

$$7! = 5040$$

The sexagesimal cycle divides it exactly:

$$5040 / 60 = 84$$

Therefore:

$$5040 = 84 \times 60$$

The four-gauge incidence witness also divides it exactly:

$$5040 / 24 = 210$$

Therefore:

$$5040 = 210 \times 24$$

The relation witness divides it exactly:

$$5040 / 30 = 168$$

Therefore:



$$5040 = 168 \times 30$$

The complete cycle admits three exact bounded readings:

84 cycles of 60  
210 blocks of 24  
168 blocks of 30

These are arithmetic decompositions of the same 7! cycle.

Each belongs to a different interpretive surface.

---

## 22. The Three Clocks

The Polyharmonic Governor organizes three computational clocks.

### 22.1 Atomic Logic Clock

$$4y^2$$

The Atomic Logic Clock governs:

nibble operations  
local carry  
binary cell transitions  
low-plane exactness

### 22.2 Spectral Observer Clock

$$16xy$$

The Spectral Observer Clock governs:

frame comparison  
observer position  
bridge relations  
cross-plane interpretation

## 22.3 Cosmic Orbit Clock

$60x^2$

The Cosmic Orbit Clock governs:

periodic phase  
block orbit  
sexagesimal scheduling  
high-plane cycle

These names describe computational roles.

They are not claims about physical time.

---

## 23. The Five Governor Modes

The Tetragrammatron Polyharmonic Governor uses five ordered modes:

|             |          |
|-------------|----------|
| FACTS       | $p = -1$ |
| RULES       | $p = 0$  |
| CLOSURES    | $p = 1$  |
| COMBINATORS | $p = 2$  |
| CONS        | $p = 3$  |

Their canonical readings are:

FACTS ground.  
RULES permit.  
CLOSURES seal.  
COMBINATORS compose.  
CONS embodies.

The governor order is consecutive:

-1, 0, 1, 2, 3

RULES occupies the Genesis/equality pivot:

$p = 0$

FACTS and CONS are circular inverse endpoints:

FACTS  
inverse grounding projection  
  
CONS  
forward embodied projection

The five modes are not five separate clocks.

They are five ways of governing the same declared relation.

---

## 24. The Binary Quadratic Form

The OMI binary quadratic form is:

$$Q(x,y) = 60x^2 + 16xy + 4y^2$$

It separates the three computational planes:

$4y^2$   
Atomic Logic Clock  
  
 $16xy$   
Spectral Observer / Bridge Plane  
  
 $60x^2$   
Cosmic Orbit Clock

The correct factor extraction is:

$$60x^2 + 16xy + 4y^2$$

$$= 4(15x^2 + 4xy + y^2)$$

It does not factor as:

$$4(3x + y)(5x + y)$$

because:

$$4(3x + y)(5x + y)$$

$$= 4(15x^2 + 8xy + y^2)$$

$$= 60x^2 + 32xy + 4y^2$$

The cross-term would be incorrect.

---

## 25. Discriminant and Positive Definiteness

For:

$$Q(x, y) = 60x^2 + 16xy + 4y^2$$

the discriminant is:

$$\Delta = b^2 - 4ac$$

with:

$$a = 60$$

$$b = 16$$

$$c = 4$$

Therefore:

$$\Delta = 16^2 - 4(60)(4)$$

$$= 256 - 960$$

$$= -704$$

Since:

```
a > 0  
 $\Delta < 0$ 
```

the form is positive definite over the real plane.

Correct statement:

```
Q(x,y) is positive definite.  
  
It has no real linear factorization.  
  
Its roots belong to a complex/projective interpretation,  
not to real transition lines.
```

Boundary:

```
The quadratic form organizes computational plane weights.  
  
It does not by itself prove a physical geometry.
```

---

## 26. Runtime Witness Structure

A minimal implementation may store the Tetragrammatron state as:

```
typedef struct {  
    uint8_t incidence_witness;    /* 0x18 */  
    uint8_t relation_witness;    /* 0x1E */  
    uint8_t centered_fold;       /* 0x3C */  
    uint8_t shifted_fold;        /* 0x78 */  
  
    uint32_t fs;                  /* 0x0001 */  
    uint32_t gs;                  /* 0x0010 */  
    uint32_t rs;                  /* 0x0100 */  
    uint32_t us;                  /* 0x1000 */  
    uint32_t gauge_carry;         /* 0x10000 */  
} TetragrammatronWitness;
```

Required invariants:

```
incidence_witness == 0x18
relation_witness  == 0x1E
centered_fold     == 0x3C
shifted_fold      == 0x78

fs == 0x0001
gs == fs << 4
rs == gs << 4
us == rs << 4
gauge_carry == us << 4
```

## 27. Runtime Adjudication Functions

The Tetragrammatron should expose separate checks:

```
bool tetragrammatron_validate_incidence(void);
bool tetragrammatron_validate_gauge_order(void);
bool tetragrammatron_validate_gauge_carry(void);

bool tetragrammatron_validate_diagonal_xor(void);
bool tetragrammatron_validate_diagonal_sum(void);
bool tetragrammatron_validate_centered_fold(void);
bool tetragrammatron_validate_shifted_fold(void);
```

The checks should retain separate provenance.

For example:

```
incidence valid
does not imply
relation closure valid

relation closure valid
does not imply
gauge incidence valid
```

A complete candidate may require both:

```
valid incidence
AND
valid relation closure
```

but the receipt should record the two results independently.

---

## 28. Receipt Form

A receipt may record:

```
{
  "incidence": {
    "valid": true,
    "witness": "0x18",
    "flags": 24,
    "gauge_order": [
      "0x0001",
      "0x0010",
      "0x0100",
      "0x1000"
    ],
    "carry": "0x10000"
  },
  "relation": {
    "valid": true,
    "diagonal_xor": {
      "d_plus": "0x00",
      "d_minus": "0x00"
    },
    "diagonal_sum": {
      "d_plus": "0x1E",
      "d_minus": "0x1E"
    },
    "centered_fold": "0x3C",
    "shifted_fold": "0x78"
  },
  "accepted": true
}
```

This preserves the authority split inside one adjudication record.

---

## 29. OMI-ISA Witness Separation

The executable witness families remain distinct.

### Null Ring

```
0x00  
0x20  
0x7F  
0xFF
```

Role:

```
byte-ring closure  
null/readable/delete/full horizon
```

### Tetragrammatron Incidence

```
0x18  
0x0001  
0x0010  
0x0100  
0x1000  
0x10000
```

Role:

```
gauge incidence  
open/sealed polarity  
hexadecimal-place stepping  
gauge carry
```

### Tetragrammatron Relation

```
0x1E  
0x3C  
0x78
```

Role:



diagonal closure  
centered fold  
shifted/full-system fold

## Metatron

0xAA55  
0x55AA

Role:

alternating balanced witness  
contrast  
scribe-facing projection

Canonical lock:

0x18 belongs to Tetragrammatron incidence.  
  
0x1E / 0x3C / 0x78 belong to Tetragrammatron relation governance.  
  
0xAA55 / 0x55AA belong to Metatron.  
  
0x00 / 0x20 / 0x7F / 0xFF belong to the Null Ring.

---

## 30. Complete Operational Sequence

The complete adjudication sequence is:

1. Omicron identifies the bounded gauge position.
2. Omnicron resolves the active circular plane.
3. OMI-Lisp declares a lazy candidate relation.
4. OMI cites the relation.
5. The Incidence Controller checks:  
FS/GS/RS/US placement,

open/sealed polarity,  
pairwise incidence,  
24-flag witness,  
hexadecimal-place carry.

6. The Relation Governor checks:

D<sup>+</sup> XOR closure,  
D<sup>-</sup> XOR closure,  
equal 0x1E sums,  
0x3C centered fold,  
0x78 shifted fold.

7. Tetragrammatron produces independent incidence and relation results.

8. Receipt records accepted or rejected state.

9. Metatron projects only accepted state.

10. IMO carries the resulting bytes, files, events, or surfaces.

---

## 31. Canonical One-Line Statement

The Tetragrammatron Incidence Controller adjudicates the ordered hexadecimal gauges `FS` `<<4 GS <<4 RS <<4 US <<4 CARRY`, preserving the `0x18` Miquel incidence witness across open/sealed gauge polarity, while the Tetragrammatron Relation Governor adjudicates the zero-XOR Polybius diagonals and their `0x1E → 0x3C → 0x78` closure-and-fold ladder; both witnesses cooperate inside one deterministic authority without conflating gauge placement, relation closure, projection, or receipt.

---

## 32. Final Canon Lock

The Tetragrammatron has two adjudicating roles.

The Incidence Controller governs:

FS = 0x0001  
GS = 0x0010  
RS = 0x0100  
US = 0x1000

The gauges advance by one hexadecimal place:

FS <<<sub>4</sub> GS <<<sub>4</sub> RS <<<sub>4</sub> US <<<sub>4</sub> CARRY

The gauge carry is:

0x10000

The four-gauge incidence witness is:

0x18 = 24 = 4!

The Relation Governor governs:

$D^+ = \{0, 5, A, F\}$

$D^- = \{3, 6, 9, C\}$

Both diagonals XOR to zero.

Both diagonals sum to:

0x1E = 30

The relation fold continues:

0x1E

→ 0x3C

→ 0x78

0x18 is incidence.

0x1E is relation.

0x10000 is hexadecimal gauge carry.

0x78 is relation fold-carry.

The witnesses cooperate.

They do not collapse.

Tetragrammatron validates.

Receipt records.

Metatron projects.

IMO carries.